

Gonzalo N. Molina

Ph.D. student in Condensed Matter Physics

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PROFILE

Computational physicist with a B.Sc. in Physics (5-year degree) from the Universidad Nacional de Río Cuarto (Argentina) and currently a Ph.D. student in Condensed Matter Physics at the Universidad Autónoma de Madrid under the supervision of Dr. José Ángel Silva-Guillén and Dr. Francisco Guinea at IMDEA Nanociencia. My research focuses on the theoretical study of two-dimensional materials and quantum systems using first-principles methods (DFT/DFPT), with applications to topological materials, transition-metal dichalcogenides, and other low-dimensional systems.

TECHNICAL SKILLS

DFT Codes Quantum ESPRESSO, SIESTA, VASP
Programming Python (NumPy, Matplotlib, ASE), Mathematica
Tools L^AT_EX, Git, Linux

RESEARCH EXPERIENCE

Predoctoral Researcher 03/2024 – Present
IMDEA Nanociencia | Madrid, Spain

- First-principles modeling of quantum and two-dimensional materials using Quantum ESPRESSO and SIESTA.

Research Stay 09/2025 – 10/2025
Departamento de Física, Universidad Nacional de Río Cuarto | Argentina

- DFT simulations of graphene and fluorinated carbon oxides for the protection of lithium metal anodes in next-generation batteries.

EDUCATION

Ph.D. in Condensed Matter Physics, Nanoscience and Biophysics 2024 – Present
Universidad Autónoma de Madrid

B.Sc. in Physics 2017 – 2023
Universidad Nacional de Río Cuarto, Argentina

PUBLICATIONS

“First-principles insights into N₂H₄-assisted exfoliation of Cu_xTiSe₂.” *Phys. Rev. B* (2026). DOI: 10.1103/w827-tsdd. [Author]

“Microscopic Insights into Magnetic Warping and Time-Reversal Symmetry Breaking in Topological Surface States of Rare-Earth-Doped Bi₂Te₃.” *Advanced Materials* (2025). DOI: 10.1002/adma.202510877. [Co-author]

CONFERENCE PRESENTATIONS

XL Reunión Bienal de la Real Sociedad Española de Física 07/2026
First-principles calculations of magnetic defects in rare-earth-doped Bi₂Te₃ | [Poster]

2026 Sesión Científica Anual del Programa de Doctorado en Física de la Materia Condensada, Nanociencia y Biofísica 06/2026
First-principles calculations of magnetic defects in rare-earth-doped Bi₂Te₃ | [Poster]

Artificial Intelligence for Advanced Materials (AI4AM 2026) 05/2026
First-principles calculations of magnetic defects in rare-earth-doped Bi₂Te₃ | [Poster]

15th Early Stage Researchers Workshop in Nanoscience 12/2025
First-principles calculations of magnetic defects in rare-earth-doped Bi₂Te₃ | [Poster]

III Workshop en Energías Renovables (UNC–UNCA–UNDEF) 10/2025
Simulación DFT de grafeno y óxidos de carbono fluorados para la protección de litio metálico en baterías | [Poster]

110° Reunión de la Asociación Física Argentina – 3ª Reunión Conjunta AFA–SUF <i>Estudio teórico de la dispersión de fonones del hBN mediante un modelo continuo</i> [Oral presentation]	09/2025
2025 Sesión Científica Anual del Programa de Doctorado en Física de la Materia Condensada, Nanociencia y Biofísica <i>Phonon Dispersion in Hexagonal Boron Nitride: A Continuum Model Approach</i> [Poster]	05/2025
22nd International Workshop on Computational Physics & Materials Science <i>Study of the exfoliation of two-dimensional Cu-doped TiSe₂</i> [Poster]	01/2025
14th Early Stage Researchers Workshop in Nanoscience <i>Phonon Dispersion in Twisted h-BN: A Continuum Model Approach</i> [Poster]	12/2024
XXXIX Reunión Bienal de la Real Sociedad Española de Física <i>Study of the exfoliation of two-dimensional Cu-doped TiSe₂</i> [Poster]	06/2024
108° Reunión Anual de Física Argentina (RAFA 2023) <i>Estudio de la exfoliación de TiSe₂ bidimensional dopado con Cu</i> [Poster]	09/2023
II Workshop en Energías Renovables (UNC–UNCA–UNDEF) <i>Óxidos de carbono fluorados para protección de Li metálico mediante DFT</i> [Poster]	10/2023
I SiModAr: Simposio de Modelado Multiescala para Biociencias y Nanomateriales en Argentina <i>Estudio mediante cálculos de primeros principios de materiales carbonosos modificados con flúor para su utilización como protección de litio metálico</i> [Poster]	11/2022
NANO 2022: XXI Encuentro de Superficies y Materiales Nanoestructurados <i>Estudio de la exfoliación de TiSe₂ dopado con Cu mediante cálculos de DFT</i> [Poster]	08/2022

SCIENCE OUTREACH

XV Feria “Madrid es Ciencia” Representative at the joint IMDEA Institutes stand, participating in nanoscience outreach activities for the general public.	2026
Semana de la Ciencia y la Innovación 2025 Workshop “Las formas del carbono: taller 3D con moléculas y efecto moiré” for children.	2025
“Investigación Computacional en Física: DFT. Del grado en la UNRC al doctorado en Madrid.” Outreach talk delivered to students and faculty members of the Departamento de Física, Universidad Nacional de Río Cuarto.	2025
“¿Cómo funciona un mar de electrones?” Oral presentation delivered as part of the lecture series “Física: Desde Átomos Hasta Volcanes”, aimed at the Facultad de Ciencias Exactas, Físico-Químicas y Naturales, Universidad Nacional de Río Cuarto.	2022

GRADUATE COURSES

Computational Spectroscopy: Hands-on Many-Body Calculations with the Yambo Code MaX CoE in collaboration with NCC Czechia.	05/2026
MaX School: Materials and Molecular Modelling with Quantum ESPRESSO MaX CoE in collaboration with NCC Czechia, NCC Austria, NCC Slovakia, NCC Slovenia, NCC Poland and NCC Hungary.	06/2024
PWTK–2024: An Online Tutorial MaX CoE in collaboration with NCC Slovenia.	05/2024
Introducción a los Métodos de Modelado Computacional en Ciencias de los Materiales Universidad Nacional de La Plata, Argentina.	10–12/2023